FactO – The Truth Engine

Course: Project Management

Report Submission

Submitted To

Md. Muhyminul Haque
Lecturer, Dept. of CSE

Submitted By

Group – 03

Rafiuzzaman Sadat – 0112330543
Najif Jawoad Jim – 0112330545
Mashurul Haque Ibtid – 0112330547
Safyal Muznabin Ratul – 0112330697

Contents

1 Introduction **1**

1.1 Project Description 1

1.2 Problem Statement 1

1.3 Motivation 2

2 Vision and Scope Document **2**

2.1 Vision Document 2

2.1.1 Project Overview 2

2.1.2 Project Purpose 2

2.1.3 Stakeholders 3

2.1.4 Limitations 3

2.2 Scope Document 3

2.2.1 In-Scope 3

2.2.2 Out-of-Scope 4

2.2.3 Functional Requirements 4

2.2.4 Non-Functional Requirements 4

3 Methodology **4**

3.1 Definition of Methodology 4

3.2 Overview of Software Development Methodologies 5

3.2.1 Waterfall Model 5

3.2.2 Prototyping Model 6

3.2.3 Agile Methodology 7

3.2.4 Scrum Framework 8

3.2.5 Spiral Model 9

3.3 Selected Methodology for FactO 10

3.4 Justification for Rejecting Alternative Methodologies 10

4 Work Breakdown Structure (WBS) **11**

4.1 Introduction 11

4.2 WBS Principles 11

4.3 Relevant Work Breakdown Structure Rules 12

4.4 WBS of FactO 13

5 Risk Management **14**

5.1 What is Risk Management 14

5.2 Risk Mitigation, Monitoring, and Management (RMMM) 14

5.2.1 Risk Mitigation 14

5.2.2 Risk Monitoring 15

5.2.3 Risk Management 15

5.3 Risk Management Plan 16

5.4 Individual Risk Information Sheets 18
5.4.1 Risk Information Sheet — Deepfake Forensics False Labels 18
5.4.2 Risk Information Sheet — Crowd Sybil Attacks & Gaming 18
5.4.3 Risk Information Sheet — Scraper Rate Limiting & Blocking 19
5.4.4 Risk Information Sheet — Legal & Defamation Liability 20
5.4.5 Risk Information Sheet — B2B Integration SLA Breach 21

6 Wideband Delphi Technique **22**
6.1 What is Wideband Delphi? 22
6.2 Why Wideband Delphi is Used for FactO 22
6.3 Individual Estimation Forms 22
6.3.1 Individual Estimation Form — Rafiuzzaman Sadat (Backend & Scraping Lead) . 23
6.3.2 Individual Estimation Form — Safyal Muznabin Ratul (AI Forensics & UI Lead) 23
6.3.3 Individual Estimation Form — Mashurul Haque Ibtid (Algorithms & B2B Lead) 24
6.3.4 Individual Estimation Form — Najif Jawoad Jim (Infrastructure & Audit Lead) . 24
6.4 Convergence Chart of Duration 24
6.5 Tentative Convergence Chart 25

Abstract

This report describes the features, capabilities, and strategic plan for our project, FactO. It explores how FactO allows users to engage easily, simplifying the misinformation reporting, evidence management, and community verification process. The analysis demonstrates how FactO could serve as a strong platform for combating digital misinformation and establishing public trust by offering insights into our future goals, system architecture, and scope of development. Ultimately, it expresses FactO's vision, ambitions, and potential effect as an innovative, real-time misinformation-response solution.

1 Introduction

1.1 Project Description

FactO is a comprehensive, web-based platform designed to support end-to-end misinformation-response workflows. The system centralizes the entire journey of fact-checking from the moment a user reports a suspicious claim to the publication of a traceable, evidence-backed verdict. Currently, misinformation verification relies on fragmented channels. FactO replaces this disjointed process with a unified, role-based application that captures claim reporting, evidence collection, community verification participation, and expert review in one place. The platform features tools for community voting, a verification log for tracking the history of a claim, and standardized "Truth Cards" to share outcomes efficiently. By bridging the gap between automated detection and human crowd-sourcing, FactO provides a Hybrid AI and Crowd verification model that operates in real-time.

1.2 Problem Statement

The digital landscape is increasingly overwhelmed by rapid cycles of misinformation. Through our benchmark analysis of existing fact-checking platforms and a targeted survey of 60 respondents, several critical gaps in the current verification ecosystem were identified:

- **Fragmented Workflows:** Misinformation handling is scattered. Users report content in one place, community discussion happens on social media, and expert verdicts are published elsewhere without a clear evidence trail.
- **Lack of Real-Time Response:** 65% of surveyed users demand reliable answers within minutes of a rumor going viral. Existing platforms rely on manual editorial pipelines that cannot scale during viral spikes.
- **Absence of Community Signal Utilization:** Almost 47% of users intuitively check comments to verify shocking posts. Existing platforms fail to harness this natural "crowd-checking" behavior into a structured verification signal.
- **Weak Transparency and Trust Indicators:** Transparency is the number one driver of trust (41.7%). However, current solutions often keep verification logs internal and lack consistent, user-facing credibility scores.
- **High-Friction Consumption:** Verification outcomes are frequently published as long, text-heavy articles rather than quick, easily shareable summaries.

1.3 Motivation

The primary motivation behind FactO is to restore trust and speed to the information verification process. With 88% of surveyed individuals expressing concern about fake or misleading information, there is a clear, urgent need for a dedicated verification model.
Our motivation is driven by the desire to:

- **Empower the Public:** Lower the barrier to reporting misinformation by providing a structured, intuitive dashboard for claim submission.
- **Democratize Verification:** Transform passive social media consumption into active community participation through structured voting and gamified engagement layers.
- **Ensure Traceability:** Provide transparent verification logs so users can see exactly how a claim was evaluated, what evidence was considered, and who made the decision.
- **Accelerate Truth Delivery:** Use "Truth Cards" to package verified outcomes into simple, highly shareable formats that can travel just as fast as the misinformation they correct.

By addressing these core needs, FactO aims to create a more resilient digital community capable of neutralizing misinformation through speed, transparency, and collaborative effort.

2 Vision and Scope Document

2.1 Vision Document

2.1.1 Project Overview

FactO is a real-time, web-based misinformation response platform engineered for the general public, expert fact-checkers, and system administrators. The primary purpose of FactO is to centralize and streamline the fragmented life cycle of misinformation handling into a unified digital ecosystem. It accomplishes this by guiding a claim from initial public reporting through evidence aggregation and community verification, culminating in an expert-reviewed, shareable outcome. In brief, the system features structured claim intake forms, evidence management tools, community verdict voting with confidence tracking, an immutable verification timeline log, standardized "Truth Card" output generation, dedicated Expert and Admin portals, gamified user rewards, and public safety demo journeys. FactO is built using an iterative development process utilizing React, Type Script, Vite, and Tailwind CSS.

2.1.2 Project Purpose

The core work of FactO is to solve the pervasive problem of fragmented, slow, and non-transparent fact-checking workflows that fail to keep pace with viral rumors. Current methods rely heavily on manual editorial processes, scattered social media discussions, and non-standardized verdict formats. FactO improves upon this baseline by introducing a real-time, hybrid verification architecture combining crowd signals with domain expert review. It dramatically increases response velocity by converting natural user "crowd-checking" behavior into structured verification data. Furthermore, it enhances public trust by providing fully traceable timeline logs detailing every piece of evidence and decision step, while publishing concise, visual "Truth Cards" designed for immediate social media dissemination.

2.1.3 Stakeholders

Primary Stakeholders:

- **Registered Users / Reporters:** Submit claims, attach evidence, and track submission progress.
- **Community Verifiers:** Active participants who evaluate evidence and submit verdict voting signals.
- **Expert Reviewers / Fact-Checkers:** Domain experts who manage review queues, evaluate evidence, and publish final outcomes.
- **Administrators:** Governance managers who oversee user roles, audit logs, reported content, and system configurations.

Secondary Stakeholders:

- **Guest / Public Visitors:** Unauthenticated users who browse public news feeds, view published Truth Cards, and access educational content.
- **Product Evaluators / Instructors:** Academic stakeholders who assess system design, implementation quality, and project requirements.

2.1.4 Limitations

- **Time Limitations:** Strict production launch windows (1 month) and regulatory compliance cycles mandate staged regional deployment phases rather than immediate global availability.
- **Budget Limitations:** Capital allocation constraints prioritize server infrastructure, security audits, and AI model inference costs over aggressive marketing campaigns.
- **Technical Limitations:** Distributed network latency and third-party platform API rate limits restrict real-time automated monitoring on high-volume external social streams.

2.2 Scope Document

2.2.1 In-Scope

- **Authentication & Access Control:** Enforce multi-factor OAuth2/JWT authentication with granular role-based permissions across public, expert, and admin scopes.
- **Claim & Evidence Pipeline:** Enable multimedia claim ingestion, metadata tracking, and secure cloud storage of supporting evidence files.
- **Hybrid Verification Engine:** Aggregate weighted user voting signals, machine learning risk scores, and expert verdicts into a definitive claim outcome.
- **Truth Card Publishing:** Generate standardized, immutable visual verdict assets optimized for web rendering and cross-platform sharing.
- **Audit Logging & Traceability:** Maintain a fully searchable transaction history recording every evidence addition, comment, and verdict state modification.
- **Notification System:** Deliver automated real-time push and email alerts regarding claim status updates and reputation reward milestones.

2.2.2 Out-of-Scope

- Fully autonomous, zero-human-in-the-loop AI adjudication for legal or high-stakes claims without human oversight.
- Legacy native desktop applications (focusing strictly on web and mobile-first responsive ecosystems).
- Direct automated content moderation or post deletion on third-party external social media platforms.
- Cryptocurrency or blockchain-backed financial payout processing for gamified community rewards.

2.2.3 Functional Requirements

- **Authentication & Access Control:** Enforce multi-factor OAuth2/JWT authentication with granular role-based permissions across public, expert, and admin scopes.
- **Claim & Evidence Pipeline:** Enable multimedia claim ingestion, metadata tracking, and secure cloud storage of supporting evidence files.
- **Hybrid Verification Engine:** Aggregate weighted user voting signals, machine learning risk scores, and expert verdicts into a definitive claim outcome.
- **Truth Card Publishing:** Generate standardized, immutable visual verdict assets optimized for web rendering and cross-platform sharing.
- **Audit Logging & Traceability:** Maintain a fully searchable transaction history recording every evidence addition, comment, and verdict state modification.
- **Notification System:** Deliver automated real-time push and email alerts regarding claim status updates and reputation reward milestones.

2.2.4 Non-Functional Requirements

- **Performance & Scalability:** Maintain sub-second API response times and support dynamic horizontal auto scaling during high-traffic viral news events.
- **Security & Data Privacy:** Comply with OWASP Top 10 standards, end-to-end data encryption at rest (AES-256) and in transit (TLS 1.3), and GDPR regulations.
- **High Availability:** Guarantee a 99.9% service up time SLA using multi-region deployment, load balancers, and automated database fail over.
- **Usability & Accessibility:** Ensure full WCAG 2.1 AA compliance and adaptive responsive UI rendering across mobile, tablet, and desktop view ports.

3 Methodology

3.1 Definition of Methodology

A software development methodology is a structured system of guidelines, procedural phases, and technical practices used to govern the life cycle of an information system. It serves as an organizational road map that defines how project requirements are gathered, designed, built, tested, and deployed, ensuring overall software quality, team alignment, and process predictability.

3.2 Overview of Software Development Methodologies

3.2.1 Waterfall Model

- Executes development in a strict linear and non-iterative sequence through predefined phases.
- Requires complete formal documentation and sign-off before progressing to the next stage.
- Emphasizes rigid gate checks where testing occurs only after the full construction phase is complete.

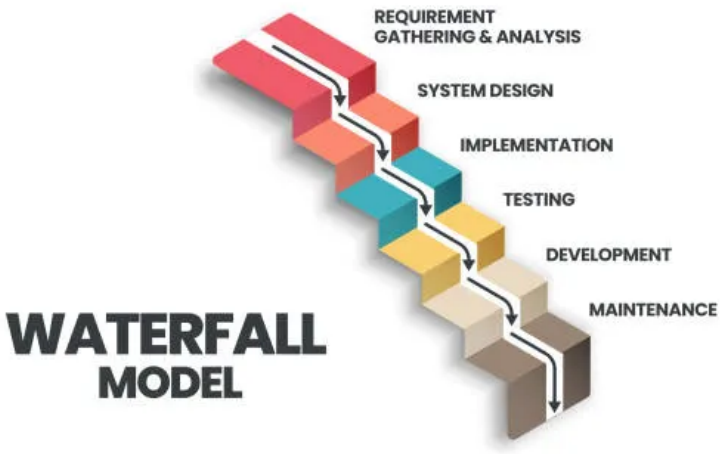


Figure 1: Sequential Phases of the Waterfall Software Development Model

Advantages:

- **Simple Structure:** Easy to understand and implement due to its straightforward, sequential phase flow.
- **Clear Milestones:** Well-defined entry and exit criteria make project status highly visible to management.
- **Disciplined Documentation:** Comprehensive written specifications ensure seamless knowledge transfer across team members.
- **Predictable Budgeting:** Fixed initial scope allows for straightforward resource allocation and cost estimation.
- **Low Operational Complexity:** Requires minimal ongoing coordination between clients and developers once initial specifications are finalized.

Disadvantages:

- **Inflexible to Change:** Extremely difficult and expensive to alter requirements once execution has begun.
- **Late Software Delivery:** Working code is not produced until late in the life cycle, leaving users unable to validate system utility early.
- **Delayed Testing Phase:** Critical bugs or design flaws are discovered late during the dedicated testing stage, leading to costly re-engineering.
- **High Failure Risk:** Misunderstandings in early requirement analysis propagate down the entire workflow undetected.

- **Poor Feedback Integration:** Lacks continuous feedback loops with stakeholders, often resulting in software that fails to meet actual user expectations.

3.2.2 Prototyping Model

- Focuses on building early, working representations (prototypes) of system features for user evaluation.
- Driven by an iterative cycle of continuous feedback collection, refinement, and re-evaluation.
- Prioritizes early graphical user interface design and interaction over backend technical depth.

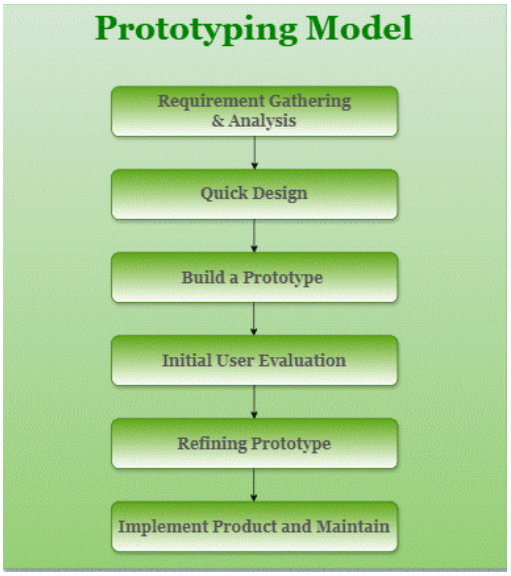


Figure 2: Iterative Prototyping Life Cycle and User Feedback Loop

Advantages:

- **Clarified Requirements:** Helps users discover missing or ambiguous needs by interacting with a tangible visual model.
- **Enhanced User Engagement:** Direct involvement fosters strong stakeholder ownership and satisfaction early in the project.
- **Early Usability Testing:** UX/UI layout flaws are exposed and resolved prior to core engineering investment.
- **Reduced Miscommunication:** Visual mocks eliminate subjective misinterpretations between developers and non-technical clients.
- **Reduced Risk of Rejection:** Continuous refinement minimizes the risk of producing an end product that users refuse to adopt.

Disadvantages:

- **High Scope Creep:** Frequent client reviews often result in endless requests for additional features and modifications.
- **Poor Architectural Quality:** Quick prototype implementations are frequently patched together, leading to substandard code base foundations if reused.
- **Unrealistic Client Expectations:** Stakeholders may mistake a non-functional front end prototype for a

nearly complete software product.

- **Increased Cost and Effort:** Multiple iteration loops can burn through allocated budget and extend timelines significantly.
- **Documentation Neglect:** Fast-paced iterations often leave the development team without formal technical documentation.

3.2.3 Agile Methodology

- Structures development into short, incremental releases focused on delivering functional software value.
- Prioritizes cross-functional collaboration and flexible adaptability over rigid planning.
- Integrates continuous stakeholder feedback throughout every phase of the project life cycle.

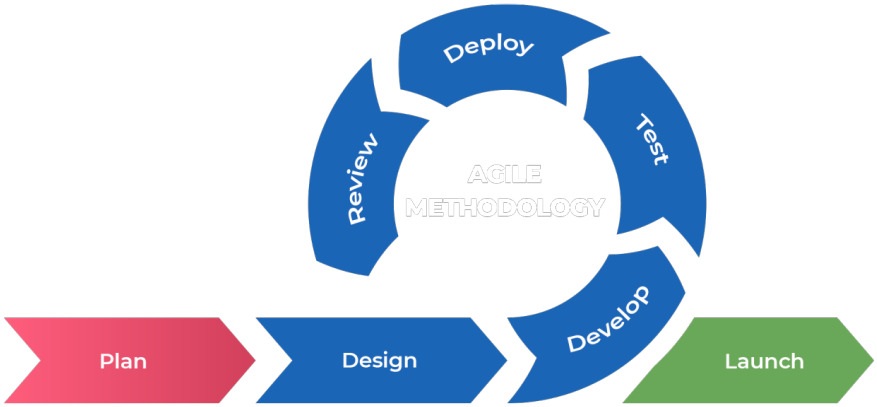


Figure 3: Iterative and Incremental Agile Development Life cycle

Advantages:

- **High Adaptability:** Accommodates changing market conditions and user demands effortlessly throughout development.
- **Fast Time-to-Market:** Core functional increments are shipped early, providing immediate value to end users.
- **Continuous Quality Assurance:** Testing is conducted continuously alongside development, ensuring early defect detection.
- **Transparent Progress:** Regular releases and status updates keep stakeholders fully informed of real system capabilities.
- **Customer-Centric Focus:** Continuous involvement ensures the final application directly addresses real user requirements.

Disadvantages:

- **Unpredictable Budgeting:** Evolving requirements make long-term cost, scope, and timeline forecasting challenging.
- **High Active Commitment:** Demands constant time and involvement from business stakeholders and

product owners.

- **Risk of Scope Drift:** Without firm governance, flexible boundaries can lead to perpetual feature creep.
- **Sparse Technical Documentation:** Focus on working software can result in incomplete architectural documentation.
- **Team Dependency:** Requires disciplined, highly experienced team members who can self-organize effectively.

3.2.4 Scrum Framework

- Operates in fixed-length, time boxed work iterations called Sprints (typically lasting 1 to 4 weeks).
- Enforces explicit team roles: Product Owner, Scrum Master, and cross-functional Development Team.
- Employs structured ceremonial meetings: Sprint Planning, Daily Stand ups, Sprint Reviews, and Retrospectives.

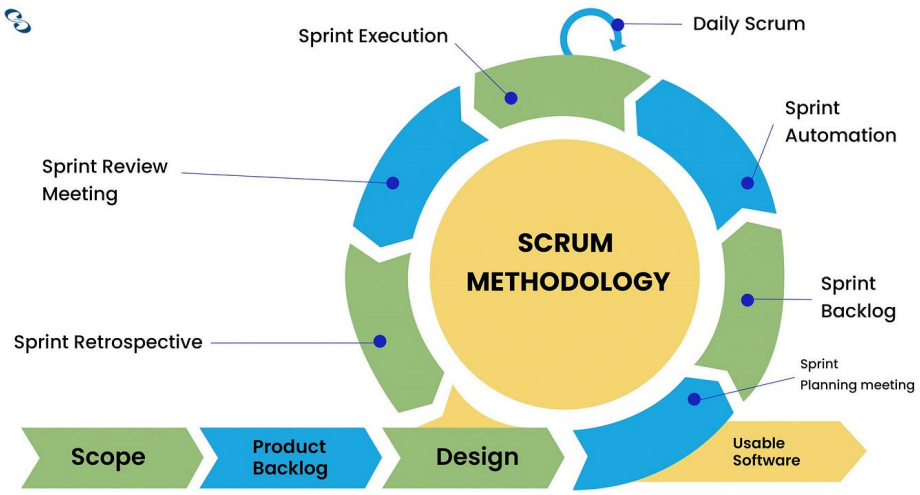


Figure 4: The Scrum Framework Workflow, Roles, and Ceremonies

Advantages:

- **Increased Productivity:** Time boxed Sprints create focus, urgency, and high operational throughput.
- **Rapid Issue Resolution:** Daily stand up meetings expose impediments immediately, preventing workflow bottlenecks.
- **Clear Operational Visibility:** Backlogs and task boards provide transparent tracking of every team task.
- **Empowered Ownership:** Self-organizing structure boosts team morale, accountability, and individual ownership.
- **Continuous Improvement:** Retrospectives allow teams to refine their internal processes continuously after every Sprint.

Disadvantages:

- **High Meeting Overhead:** Mandatory ceremonies can feel disruptive and consume substantial produc-

tive coding time.

- **Vulnerability to Scope Creep:** Unrefined product backlogs can allow unvetted tasks to bloat active Sprints.
- **Critical Role Dependency:** Poor performance by a Scrum Master or Product Owner severely degrades team output.
- **Scaling Difficulty:** Standard Scrum can struggle to coordinate across large, multi-team enterprise environments without add-on frameworks.
- **Team Burnout Risk:** The relentless pace of back-to-back Sprints can exhaust developers over long periods.

3.2.5 Spiral Model

- Combines the structured, systematic nature of Waterfall with the iterative approach of Prototyping.
- Driven fundamentally by continuous, rigorous risk analysis at every execution phase.
- Progresses outward through four main quadrant loops: Planning, Risk Analysis, Engineering, and Evaluation.

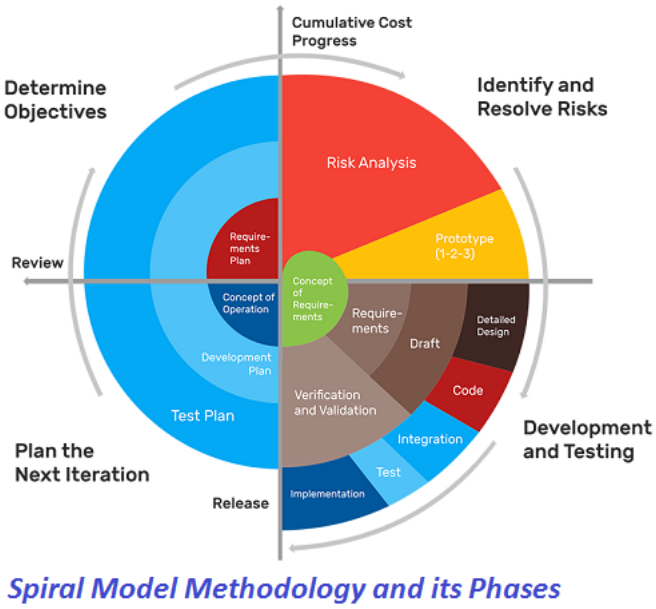


Figure 5: Quadrant Cycles and Continuous Risk Assessment in the Spiral Model

Advantages:

- **Superior Risk Management:** Built-in, expert risk analysis prevents catastrophic project failures.
- **Suitable for Complex Systems:** Designed explicitly for large-scale, mission-critical enterprise engineering.
- **Flexible Requirement Accommodation:** Modifications can be seamlessly built into future outer spiral loops.
- **High Quality and Security:** Continuous evaluations ensure rigorous software reliability and security standards.

- **Early Life cycle Prototyping:** Uses early functional models to validate critical technical assumptions.

Disadvantages:

- **Extremely High Cost:** Intensive risk assessment and continuous loops make it financially prohibitive for small projects.
- **High Operational Complexity:** Managing quadrant transitions and spiral iterations requires complex governance.
- **Heavy Expertise Reliance:** The success of the project relies almost entirely on highly skilled risk assessment experts.
- **Excessive Documentation:** Extensive administrative overhead slows down actual software development.
- **Uncertain Completion Timelines:** Open-ended iteration loops can make final delivery dates difficult to guarantee.

3.3 Selected Methodology for FactO

For the engineering of **FactO**, we selected the **Agile Methodology utilizing a Scrum framework**. The choice of Agile/Scrum is directly aligned with the complex, multi-component architecture of our platform and the fast-paced nature of digital misinformation response:

- **Iterative UI/UX Module Delivery:** FactO's front-end is constructed using React, Type-Script, and Tailwind CSS across four distinct role-based interfaces (Public Reporter, Community Verifier, Expert Reviewer, and Admin). Scrum allows us to deliver these modular interfaces across dedicated 2-week Sprints, testing usable components—such as the claim submission modal and interactive verification timeline log—incrementally.
- **Algorithm Calibration and Feedback Loops:** The core engine of FactO relies on a hybrid verification architecture that balances weighted user voting signals, machine learning risk scores, and expert verdicts. Developing this algorithm requires continuous empirical testing; Sprint retrospectives and reviews provide the feedback loops needed to fine-tune signal-weighting formulas based on real user interactions.
- **Rapid Delivery of Core Product Assets:** Standalone features like the automated visual "Truth Card" generator and push-notification workflows can be prioritized independently in the Product Backlog, ensuring functional, high-value components are shippable early in the project life cycle.

3.4 Justification for Rejecting Alternative Methodologies

- **Why Not Waterfall:** Waterfall requires complete, immutable phase completion before development begins. In FactO, locking down system specifications upfront would prevent us from adapting the hybrid crowd-voting algorithm or user-facing Truth Card formats based on empirical testing. Furthermore, waiting until a final testing phase to evaluate the complex integration of multi-factor OAuth2/JWT authentication, evidence storage pipelines, and role-based portals introduces unacceptable integration risk and potential project delay.
- **Why Not Prototyping (Standalone):** While early visual mock ups are useful for UI layout, relying solely on a Prototyping model lacks the structured backlog governance and sprint cadence

needed to build production-ready backend infrastructure. FactO requires robust Non-Functional standards—including AES-256 data encryption, OWASP Top 10 security compliance, and persistent, searchable audit logging—which risk being compromised or built on fragile, patched code base foundations under a purely prototype-driven workflow.

- **Why Not Spiral:** The Spiral model is engineered for massive enterprise applications with high technical and financial risk, placing heavy emphasis on formal, multi-quadrant risk analysis before every iteration. Applying Spiral to FactO would impose excessive administrative overhead and severe timeline inflation, hindering the fast deployment velocity required for a web-based misinformation management platform.

4 Work Breakdown Structure (WBS)

4.1 Introduction

A Work Breakdown Structure (WBS) is a hierarchical method used to divide a project into smaller, manageable, and clearly defined components. It organizes the total scope of a project into different levels of tasks and activities, making the project easier to plan, schedule, assign, monitor, and control. A WBS helps project teams understand the work required, establish responsibilities, estimate resources and costs, and track project progress effectively. It also provides a structured framework for ensuring that all necessary project work is identified and included.

4.2 WBS Principles

- **Hierarchical Structure:** A WBS follows a hierarchical structure, where the project is divided into progressively smaller and more manageable components. The highest level represents the entire project, while lower levels break the project into major deliverables, sub-deliverables, and work packages. This structure makes it easier to understand the relationship between the overall project and individual pieces of work.
- **Outcome-Oriented:** Elements focus strictly on deliverables, results, or outcomes that must be produced rather than concentrating solely on individual activities. This ensures every part contributes directly to achieving project objectives.
- **100% Rule:** The 100% rule states that the WBS must include 100% of the work required to complete the project according to its defined scope. Every necessary deliverable and work component is represented, while out-of-scope work is strictly excluded to prevent scope creep.
- **Mutually Exclusive Elements:** WBS elements are mutually exclusive; each component represents a distinct piece of work without overlapping with another component, avoiding duplication of effort and responsibility.
- **4% Rule of Decomposition:** A work package should generally represent no more than 4% of the total project effort or scope. If a work package is larger, it is divided into smaller components to make it easier to estimate and monitor.
- **40-Hour Rule of Decomposition:** Ideally, a work package should require around 40 hours or less to complete. Tasks requiring significantly more time are decomposed into smaller, more manageable units.

- **8/80 Rule:** A work package should generally require at least 8 hours and no more than 80 hours of effort to complete, maintaining a practical level of detail without over-granularity or loss of control.
- **Three Levels (Level 2 Focus):** A basic WBS is organized into three levels: Level 1 (complete project), Level 2 (major deliverables/work areas), and Level 3 (work packages). Level 2 is the most vital as it establishes the main structural foundation for scheduling, budgeting, and responsibility matrix allocation.

4.3 Relevant Work Breakdown Structure Rules

- **Collaborative Creation:** The project team must collaborate to create the WBS, ensuring both project scope and product scope are clearly communicated and agreed upon.
- **Strict Subset Hierarchy:** Every lower-level WBS component is a direct subset of the higher-level component above it.
- **Flexible Decomposition Depth:** Different branches of the WBS may decompose to varying levels of depth based on task complexity.
- **Deliverable-Driven Focus:** Prioritize tangible results and outcomes over continuous operational actions or vague activities.
- **Essential Deliverables Only:** Exclude unnecessary tasks or administrative bloat that do not directly contribute to the project scope.
- **WBS Dictionary Support:** Work packages should be clearly and concisely described in a WBS dictionary, using noun-oriented naming conventions.
- **Standardized Numbering System:** Standardize level numbering after structural definition (e.g., 1, 1.1, 1.1.1) to maintain traceability.
- **Explicit Ownership Assignment:** Assign clear responsibility to a specific team member or owner for every individual work package.

4.4 WBS of FactO

Figure 6: Work Breakdown Structure (WBS) of FactO – The Truth Engine

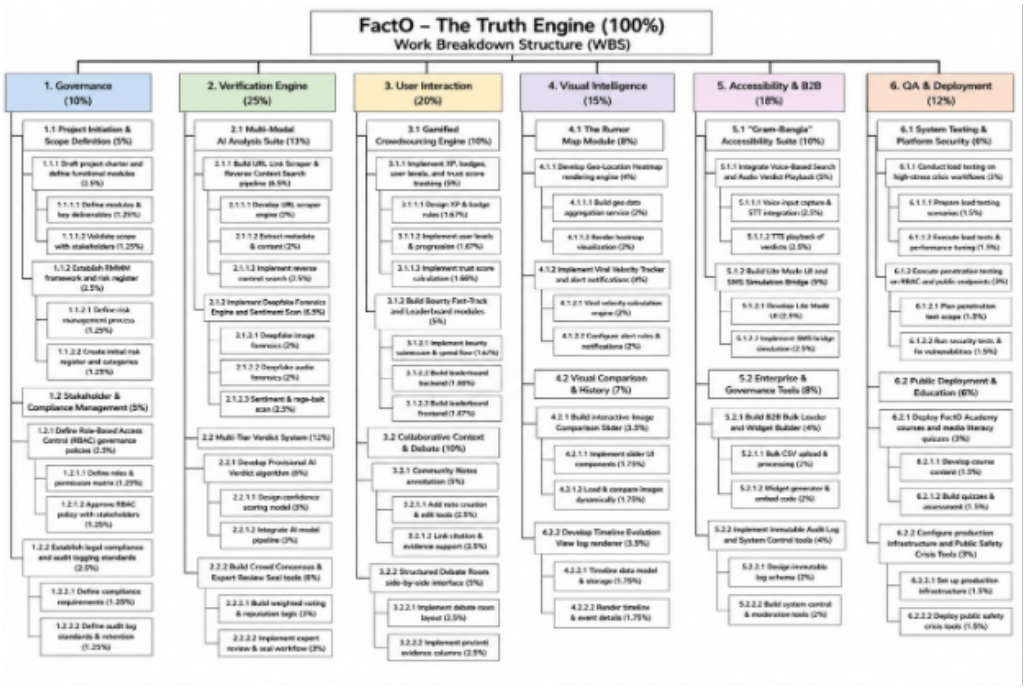


Table 1: Task Schedule and Cost Allocation of FactO Modules

Task Title	Owner	Start Date	Due Date	Dur. (Months)	Cost ($)
Project Charter & Functional Modules	Sadat	01/05/2025	15/05/2025	0.5	625
RMMM Framework & Risk Register	Ratul	16/05/2025	31/05/2025	0.5	625
RBAC Governance Policies	Ibtid	01/06/2025	15/06/2025	0.5	625
Legal Compliance & Audit Standards	Jaowad	16/06/2025	30/06/2025	0.5	625
URL Scraper & Reverse Context Search	Sadat	01/07/2025	15/08/2025	1.5	1,625
Deepfake Forensics & Sentiment Scan	Ratul	16/08/2025	30/09/2025	1.5	1,625
Provisional AI Verdict Algorithm	Ibtid	01/10/2025	07/11/2025	1.25	1,500
Crowd Consensus & Expert Review	Jaowad	08/11/2025	15/12/2025	1.25	1,500
XP, Badges & Trust Score System	Sadat	16/12/2025	15/01/2026	1.0	1,250
Bounty Fast-Track & Leaderboard	Ratul	16/01/2026	15/02/2026	1.0	1,250
Community Notes System	Ibtid	16/02/2026	07/03/2026	0.75	1,250
Structured Debate Room	Jaowad	08/03/2026	31/03/2026	0.75	1,250
Geo-Location Heatmap Engine	Sadat	01/04/2026	22/04/2026	0.75	1,000
Viral Velocity Tracker & Alerts	Ratul	23/04/2026	15/05/2026	0.75	1,000
Interactive Image Comparison Slider	Ibtid	16/05/2026	07/06/2026	0.75	875
Timeline Evolution View	Jaowad	08/06/2026	30/06/2026	0.75	875
Voice Search & Audio Verdict Playback	Sadat	01/07/2026	31/07/2026	1.0	1,250
Lite Mode UI & SMS Simulation Bridge	Ratul	01/08/2026	31/08/2026	1.0	1,250
B2B Bulk Loader & Widget Builder	Ibtid	01/09/2026	30/09/2026	1.0	1,000
Immutable Audit Log & System Control	Jaowad	01/10/2026	31/10/2026	1.0	1,000
Load Testing & Crisis Workflow Testing	Sadat	01/11/2026	15/11/2026	0.5	750
Penetration Testing of RBAC & Endpoints	Ratul	16/11/2026	30/11/2026	0.5	750
FactO Academy & Media Literacy Quizzes	Ibtid	01/12/2026	15/12/2026	0.5	750
Production Infrastructure & Crisis Tools	Jaowad	16/12/2026	31/12/2026	0.5	750
Total Estimation				20 Months	$25,000

5 Risk Management

5.1 What is Risk Management

Risk management is the systematic process of identifying, analyzing, evaluating, monitoring, and controlling possible problems or uncertain events that may negatively affect a project's objectives. Every software project carries inherent risks that can cause schedule delays, cost overruns, quality degradation, security vulnerabilities, or total project failure. Risk management enables the project team to uncover these potential bottlenecks early and prepare suitable mitigation plans to reduce their probability and impact.

In the case of **FactO – The Truth Engine**, risk management is particularly vital given that the platform is a complex misinformation verification system combining AI multi-modal analysis, crowd sourcing consensus, expert review workflows, accessibility services, visual intelligence, and strict administrative controls. Consequently, FactO faces specialized risks such as incorrect AI-generated verdicts, failures in detecting novel deepfakes, community vote manipulation (Sybil attacks), unauthorized platform access, performance degradation during traffic surges, and legal liability stemming from sensitive claims. Systematic risk management ensures that FactO remains accurate, secure, reliable, scalable, transparent, and trustworthy throughout its development life cycle.

5.2 Risk Mitigation, Monitoring, and Management (RMMM)

Risk Mitigation, Monitoring, and Management (RMMM) is an operational framework used to manage project risks continuously across the development cycle. It equips the project team to anticipate potential failures, continuously track risk indicators during execution, and execute targeted contingency plans if risks materialize.

5.2.1 Risk Mitigation

Risk mitigation encompasses preemptive strategies executed before a risk materializes to lower its probability or lessen its severity:

- **Incorrect AI Verdicts:** Implement a hybrid verification architecture combining machine learning scores with weighted crowd consensus and mandatory human expert sign-off.
- **Deepfake Detection Failures:** Continuously update synthetic datasets and retrain the Deepfake Forensics Engine against emerging video and audio synthesis techniques.
- **Crowd Vote Manipulation:** Enforce weighted voting algorithms tied to user trust levels, reputation scores, historical accuracy, and device fingerprinting.
- **Unauthorized Access:** Enforce strict Role-Based Access Control (RBAC), multi-factor authentication (MFA), and token-based OAuth2/JWT governance.
- **Security Vulnerabilities:** Conduct routine dynamic security audits, automated dependency vulnerability scans, and rigorous penetration testing.
- **System Scalability Bottlenecks:** Execute high-stress load testing and architect backend pipelines with auto-scaling microservices and CDN edge caching.
- **Low Community Engagement:** Integrate gamification elements including XP rewards, badges, trust levels, leaderboards, and structured Jury Duty tasks.

- **Accessibility Service Failures:** Regularly test Gram-Bangla accessibility suites and maintain graceful fallback options (e.g., Lite Mode text rendering).
- **Legal & Sensitive Content Liability:** Publish clear disclaimers distinguishing provisional crowd data from binding Expert Seals, supported by detailed audit logging.
- **High Structural Complexity:** Deconstruct FactO into isolated, modular work components (Engine, Interaction, Intelligence, Accessibility) developed and tested independently.

5.2.2 Risk Monitoring

Risk monitoring tracks risk indicators continuously across operational phases to detect early warning signs:

- **AI Accuracy Monitoring:** Continuously compare AI provisional scores against final expert verdicts to log false positive/negative rates.
- **Deepfake Engine Tracking:** Monitor forensics precision metrics specifically when handling novel generative formats or low-resolution uploaded media.
- **Crowd Anomaly Detection:** Monitor vote ingestion logs for sudden volume spikes, IP clustering, or coordinated voting patterns indicating Sybil manipulation.
- **System Performance Metrics:** Real-time tracking of server latency, CPU/RAM utilization, database connection pools, and query processing times.
- **Security Event Audit:** Log failed login attempts, privilege escalation queries, API abuse, and unauthenticated administrative endpoint calls.
- **User Reputation Auditing:** Track rapid fluctuations in user trust scores and flag accounts demonstrating suspicious voting accuracy shifts.
- **Accessibility Pipeline Health:** Monitor latency and error rates across Speech-to-Text (STT), Text-to-Speech (TTS), and SMS bridge connections.
- **Community Participation Analytics:** Track active verifier counts, community note contributions, and leaderboard completion rates.
- **Audit Log Verification:** Periodically review the immutable event stream to confirm system transparency and detect unauthorized state changes.
- **Schedule and Budget Velocity:** Track Sprint burn-down charts, task completion milestones, and resource utilization to catch schedule slip early.

5.2.3 Risk Management

Risk management dictates reactive actions to contain damage when an identified risk occurs:

- **AI Verdict Correction:** Automatically flag disputed AI ratings, reset provisional status, and route the claim to high-priority Expert Review.
- **Deepfake Fallback Handling:** Manually reassign unclassifiable synthetic media to specialized forensic teams while incorporating failed inputs into retraining pipelines.
- **Sybil Attack Containment:** Instantly freeze compromised user accounts, strip invalid voting weights, and recalculate crowd consensus baseline scores.

- **Security Incident Response:** Revoke compromised JWT session tokens, isolate affected endpoints, apply patch hotfixes, and reset access controls.
- **System Outage Recovery:** Trigger automated load balancing failover, spin up backup database read-replicas, and restrict non-essential frontend features.
- **Data Corruption Restoration:** Restore corrupted state entries from immutable, cryptographically hashed cold backups and log the root cause.
- **Accessibility Failover:** Route failing voice/TTS request pipelines automatically to Lite Mode UI text endpoints without interrupting user sessions.
- **Engagement Boosting:** Adjust XP multiplier incentives and highlight active bounties during periods of decreased crowdsourcing engagement.
- **Legal Content Moderation:** Swiftly apply temporary content masking under administrative authority for flagged claims under active legal disputes.
- **Schedule Realignment:** Reallocate development resources, adjust minor scope items, or run parallel micro-sprints to mitigate schedule slip.

5.3 Risk Management Plan

Assessment Team Members: *Sadat, Ratul, Ibtid, Jaowad*

ID	Risk Description	P	I	Pri	Mitigation Strategies	Responsible Person
R01	**Deepfake Forensics False Labels:** Advanced generative audio/video models outpace detection mechanisms, causing automated mislabeling.	2	3	6	1. Implement continuous retraining pipelines on fresh synthetic datasets. 2. Enforce visible confidence intervals and mandate expert review for borderline scores.	Ratul
R02	**Crowd Sybil Attacks & Gaming:** Coordinated bot networks or collusive users manipulate crowd consensus weighting.	3	3	9	1. Enforce IP/device fingerprinting and MFA for vote weighting. 2. Implement dynamic reputation decay algorithms and automated bot detection routines.	Sadat
R03	**Scraper Rate Limiting & Blocking:** External social/news platforms alter DOM structures or block IPs, failing scraper ingestion.	3	2	6	1. Deploy distributed headless scrapers with automated proxy rotation. 2. Use fallback Open Graph parsing and support manual user archive uploads.	Sadat

ID	Risk Description	P	I	Pri	Mitigation Strategies	Responsible Person
R04	**Low-Bandwidth Infrastructure Bottlenecks:** Multi-modal evidence payloads cause high latency on low-speed 2G networks.	2	3	6	1. Pre-render and cache TTS audio outputs at CDN edge locations. 2. Serve zero-JS, text-only payloads via Lite Mode UI.	Ratul
R05	**Audit Log Storage Blowout:** High-volume edit histories and vote logs create database scaling bottlenecks.	2	2	4	1. Utilize an append-only, high-throughput event ledger (Kafka). 2. Partition logs into cold storage while preserving cryptographic hash continuity.	Jaowad
R06	**Voice Engine Dialect Misinterpretation:** STT model errors on regional Bangla dialects lead to inaccurate searches.	3	2	6	1. Integrate fine-tuned regional acoustic models for Bangla speech. 2. Prompt users with a text confirmation screen prior to vector search execution.	Sadat
R07	**Legal & Defamation Liability:** Publicly exposing unverified claims triggers legal threats before expert sign-off.	2	3	6	1. Display clear disclaimers distinguishing provisional data from Expert Seals. 2. Provide rapid administrative quash pathways under strict RBAC controls.	Jaowad
R08	**Geo-Heatmap GIS Latency:** Traffic surges during viral news crises stall real-time map rendering routines.	2	2	4	1. Decouple ingress processing from rendering using spatial databases (PostGIS). 2. Serve geo-tiles via pre-cached feeds and web-socket payloads.	Sadat
R09	**B2B Integration SLA Breach:** Bulk CSV verifications from enterprise clients overload core web APIs.	1	3	3	1. Isolate B2B workloads onto dedicated worker queues with rate limits. 2. Implement asynchronous background processing decoupled from main web engine.	Ibtid
R10	**Bounty System Exploitation:** Malicious users submit low-value or spam rumors to fast-track personal agendas.	2	1	2	1. Enforce minimum reputation thresholds and non-refundable XP deposits. 2. Require algorithmic web traction validation before fast-tracking queue items.	Ratul

Table 2: Risk Plan and Mitigation Strategy for FactO (P = Probability [1–5], I = Impact [1–5], Pri = Priority [$P \times I$])

5.4 Individual Risk Information Sheets

5.4.1 Risk Information Sheet — Deepfake Forensics False Labels

Risk ID: R01
Risk: Deepfake Forensics False Labels
Category: AI / Computer Vision
Probability: 2
Impact: 3
Risk Exposure: 6

Description Advanced generative media tools outpace baseline AI detection models, causing synthetic video or audio uploads to be incorrectly tagged as authentic, or vice versa.

Refinement / Context FactO relies heavily on multi-modal AI models for deepfake forensics. If novel diffusion models or voice clones bypass detection, users receive incorrect provisional verdicts, damaging platform trust.

Mitigation

1. Implement continuous automated retraining pipelines using fresh synthetic media datasets.
2. Enforce strict confidence threshold limits; automatically tag low-confidence verdicts as ambiguous.
3. Require mandatory human expert review for borderline media artifacts.

Monitoring

1. Track false positive and false negative rates against expert post-audit conclusions.
2. Monitor confidence score distributions across all processed media uploads.
3. Review detection accuracy when new generative media tools enter the public domain.

Management / Contingency Plan

1. Instantly reset disputed deepfake confidence flags to non-binding status.
2. Route unclassifiable media assets to specialized forensic review sub-queues.
3. Ingest misclassified media back into active training pipelines for emergency model patching.

Current Status: Under Monitoring
Assigned: Safyal Muznabin Ratul

5.4.2 Risk Information Sheet — Crowd Sybil Attacks & Gaming

Risk ID: R02
Risk: Crowd Sybil Attacks & Reputation Gaming
Category: Security / Governance
Probability: 3
Impact: 3
Risk Exposure: 9

Description Coordinated bot networks or collusive groups attempt to manipulate community consensus scores by mass-voting on targeted claims to promote misinformation.

Refinement / Context FactO leverages crowdsourced trust scores and community notes to assess claims rapidly. Attackers may spin up multiple fake accounts to sway consensus decisions on politically or socially sensitive topics.

Mitigation

1. Enforce device fingerprinting, IP subnet rate limits, and multi-factor authentication for voting access.
2. Implement dynamic trust decay algorithms that weight votes according to historical accuracy and account age.
3. Deploy automated anomaly detection models to detect synchronized voting clusters.

Monitoring

1. Monitor real-time vote ingestion logs for abnormal velocity spikes on specific claims.
2. Track IP subnet clustering and rapid account creation trends.
3. Audit trust score deviations among high-volume voters.

Management / Contingency Plan

1. Instantly freeze suspicious account networks and strip their voting weight.
2. Recalculate baseline consensus scores without flagged accounts.
3. Escalate targeted claims directly to verified expert reviewers for override.

Current Status: Active Mitigation
Assigned: Rafiuzzaman Sadat

5.4.3 Risk Information Sheet — Scraper Rate Limiting & Blocking

Risk ID: R03
Risk: Scraper Rate Limiting & Target IP Blocking
Category: Integration / Data Ingestion
Probability: 3
Impact: 2
Risk Exposure: 6

Description Target news platforms, social networks, or search engines alter their DOM structures or block automated scraper IP ranges, halting automated context retrieval.

Refinement / Context The URL Scraper and Reverse Context Engine fetch external web metadata to verify source authenticity. Persistent IP blocks prevent automated ingestion of primary source data.

Mitigation

1. Deploy distributed headless scrapers operating over residential proxy rotation networks.
2. Implement resilient fallback mechanisms using OpenGraph tags and public archival APIs.
3. Allow manual user archive uploads as secondary verification inputs.

Monitoring

1. Monitor scraper success/failure rates by target domain.
2. Track HTTP status codes (403 Forbidden, 429 Too Many Requests, 503 Service Unavailable).
3. Alert maintainers when HTML DOM parser exceptions exceed defined thresholds.

Management / Contingency Plan

1. Automatically rotate blocked scraper nodes to backup proxy pools.
2. Switch failed automated requests to OpenGraph or public archive API fallbacks.
3. Prompt users to manually submit raw text or archived links when automated fetching fails.

Current Status: Under Monitoring
Assigned: Rafiuzzaman Sadat

5.4.4 Risk Information Sheet — Legal & Defamation Liability

Risk ID: R07
Risk: Legal & Defamation Liability
Category: Legal / Compliance
Probability: 2
Impact: 3
Risk Exposure: 6

Description Publicly publishing provisional ratings or community notes on disputed claims may trigger defamation actions or legal takedown notices prior to human expert sign-off.

Refinement / Context FactO rates public statements and viral claims. Labeling a statement as false based solely on early provisional crowd data exposes the organization to legal challenges if the claim is later validated.

Mitigation

1. Render clear UI disclaimers distinguishing unverified provisional crowd assessments from official Expert Seals.
2. Establish administrative content-quashing protocols under strict Role-Based Access Control (RBAC).
3. Conduct legal reviews of publishing guidelines and automated system disclaimers.

Monitoring

1. Monitor user flag logs for incoming legal or defamation notices.
2. Review provisional claims receiving high dispute volumes from verified claim owners.
3. Track response times for handling administrative review requests.

Management / Contingency Plan

1. Apply immediate temporary content masking for claims placed under formal legal dispute.
2. Fast-track legally challenged claims to expert review panels for priority audit.
3. Record all actions within the immutable audit log for compliance documentation.

Current Status: Under Review
Assigned: Najif Jawoad Jim

5.4.5 Risk Information Sheet — B2B Integration SLA Breach

Risk ID: R09
Risk: B2B Integration SLA Breach
Category: Architecture / Multi-Tenancy
Probability: 1
Impact: 3
Risk Exposure: 3

Description Bulk request uploads from enterprise clients overload core web APIs, degrading response times and violating Service Level Agreements (SLAs).

Refinement / Context Enterprise clients submit mass URLs via CSV bulk loaders for rapid verification. Unthrottled bulk processing can starve main web endpoints of server resources.

Mitigation

1. Decouple B2B bulk uploads from the primary web application using dedicated asynchronous background queues.
2. Implement multi-tenant rate limiting and priority tier queues for enterprise endpoints.
3. Scale background worker nodes independently based on queue backlogs.

Monitoring

1. Monitor B2B API endpoint response latency and request queue backlogs.
2. Track CPU and memory utilization on background queue worker pools.
3. Audit customer SLA breach alerts and rate-limit execution counts.

Management / Contingency Plan

1. Dynamically auto-scale background queue processing instances during bulk surges.
2. Enforce tenant-level batch processing throttles to protect main system resources.
3. Route enterprise batch tasks to off-peak execution windows when necessary.

Current Status: Under Monitoring
Assigned: Mashurul Haque Ibtid

6 Wideband Delphi Technique

6.1 What is Wideband Delphi?

The Wideband Delphi technique is a consensus-based estimation methodology widely used in software project management to derive accurate effort, cost, and duration estimates. Evolved from the original Delphi method developed by the RAND Corporation, the "wideband" variation introduces structured group interaction, anonymous estimation rounds, and iterative feedback sessions.

The process follows a defined multi-step workflow:

1. **Problem Presentation:** The Project Manager presents the Work Breakdown Structure (WBS) and scope document to a panel of estimators.
2. **Anonymous Estimation (Round 1):** Each expert independently estimates task durations without consulting peers, avoiding groupthink and dominance bias.
3. **Group Discussion:** The moderator compiles all estimates into a summary chart. The team anonymously discusses wide variances, hidden technical complexities, and mismatched assumptions without pressuring individuals.
4. **Iterative Re-estimation (Rounds 2 & 3):** Estimators submit revised estimates based on insights gained from group discussions.
5. **Consensus & Final Estimation:** The process repeats until the estimates converge within an acceptable tolerance margin, yielding a final baseline duration.

6.2 Why Wideband Delphi is Used for FactO

FactO – The Truth Engine presents several architectural and domain-specific challenges that make single-expert or top-down estimation techniques unreliable:

- **High Multi-Disciplinary Complexity:** FactO integrates heterogeneous technical stacks including Computer Vision (Deepfake forensics), NLP (Bangla voice/dialect processing), low-latency web scraping, spatial databases (PostGIS heatmaps), and cryptographically hashed audit trails. No single team member possesses equal depth across all modules.
- **Mitigation of Estimator Bias:** Technical leads naturally under-estimate components within their primary area of expertise while over-estimating unfamiliar subsystems. Anonymous round-based estimation neutralizes personal optimism and pessimism.
- **Uncovering Hidden Architectural Risks:** The iterative discussion rounds between Round 1 and Round 2 encourage the team to identify unstated technical dependencies, such as proxy rotation bottlenecks or micro-service queue latency, before committing to binding project schedules.
- **Team Ownership and Alignment:** Because all four core developers (Sadat, Ratul, Ibtid, Jaowad) actively participate in converging the figures, collective commitment to the final project baseline is significantly higher.

6.3 Individual Estimation Forms

The team conducted three estimation rounds across the main functional task clusters of FactO. Below are the individual estimation records for each team member.

6.3.1 Individual Estimation Form — Rafiuzzaman Sadat (Backend & Scraping Lead)

Task Cluster / Module	R1 (M)	R2 (M)	R3 (M)	Rationale for Revision
Architecture Scope & Charter	2.0	1.5	1.0	Simplified functional module specifications.
Scraper & Context Search Engine	6.5	5.5	5.0	Integrated distributed headless proxy pools.
XP, Badges & Trust Score System	4.5	4.0	3.5	Optimized trust score decay formula logic.
Geo-Location Heatmap Engine	3.5	3.0	2.5	Utilized pre-cached spatial PostGIS tiles.
Voice Search & Audio Playback	5.0	4.5	4.5	Adopted fine-tuned acoustic STT models.
Load Testing & System Integration	3.5	3.5	3.5	Standardized automated load testing scripts.
Total Project Duration Estimate	**25.0**	**22.0**	**20.0**	**Converged to 20-month total project timeline**

Table 3: Estimation Form — Rafiuzzaman Sadat

6.3.2 Individual Estimation Form — Safyal Muznabin Ratul (AI Forensics & UI Lead)

Task Cluster / Module	R1 (M)	R2 (M)	R3 (M)	Rationale for Revision
RMMM Framework & Risk Register	2.5	2.0	1.0	Consolidated project risk categories.
Deepfake Forensics Engine	7.0	6.0	5.5	Decoupled continuous retraining pipelines.
Bounty Fast-Track & Leaderboard	4.5	4.0	3.5	Streamlined leaderboard database indexing.
Viral Velocity Tracker & Alerts	3.5	3.0	2.5	Reused real-time event-streaming triggers.
Lite Mode UI & SMS Bridge	5.0	4.5	4.5	Pre-rendered static UI components.
Penetration Testing & Security	3.5	3.0	3.0	Focused endpoint security audit scope.
Total Project Duration Estimate	**26.0**	**22.5**	**20.0**	**Converged to 20-month total project timeline**

Table 4: Estimation Form — Safyal Muznabin Ratul

6.3.3 Individual Estimation Form — Mashurul Haque Ibtid (Algorithms & B2B Lead)

Task Cluster / Module	R1 (M)	R2 (M)	R3 (M)	Rationale for Revision
RBAC Governance Policies	2.0	1.5	1.0	Standardized user authorization roles.
Provisional AI Verdict Algorithm	6.0	5.5	5.0	Simplified weighted consensus calculation.
Community Notes System	4.5	4.0	3.5	Reused markdown parsing components.
Interactive Image Slider	3.5	3.0	2.5	Adopted lightweight HTML canvas library.
B2B Bulk Loader & Widgets	4.5	4.5	5.0	Offloaded batch loader to async queues.
FactO Academy Literacy Quizzes	3.5	3.0	3.0	Modularized quiz question structure.
Total Project Duration Estimate	**24.0**	**21.5**	**20.0**	**Converged to 20-month total project timeline**

Table 5: Estimation Form — Mashurul Haque Ibtid

6.3.4 Individual Estimation Form — Najif Jawoad Jim (Infrastructure & Audit Lead)

Task Cluster / Module	R1 (M)	R2 (M)	R3 (M)	Rationale for Revision
Legal Compliance & Standards	2.0	1.5	1.0	Finalized administrative legal disclaimers.
Crowd Consensus & Expert Review	5.5	5.0	5.0	Standardized human expert review workflow.
Structured Debate Room	4.0	3.5	3.5	Integrated standard websocket channels.
Timeline Evolution View	3.5	3.0	2.5	Implemented pre-rendered dynamic timelines.
Immutable Audit Log Engine	4.5	4.5	4.5	Implemented append-only event stream (Kafka).
Production Infrastructure Setup	3.5	3.5	3.5	Automated deployment via Docker scripts.
Total Project Duration Estimate	**23.0**	**21.0**	**20.0**	**Converged to 20-month total project timeline**

Table 6: Estimation Form — Najif Jawoad Jim

6.4 Convergence Chart of Duration

The estimation variance narrowed significantly over the three Wideband Delphi rounds. In Round 1, team estimates for overall project duration varied between 23.0 and 26.0 months. Through group evaluation and risk resolution in Round 2, the variance decreased substantially (ranging from 21.0 to 22.5 months), finally converging in Round 3 to an agreed baseline duration of **20.0 months** across all four core team members.

6.5 Tentative Convergence Chart

Tentative Convergence Chart

Round	Sadat	Ratul	Ibtid	Jaowad	Range (Min–Max)	Average
Round 1	25.0 months	26.0 months	24.0 months	23.0 months	23.0 – 26.0	24.5 months
Round 2	22.0 months	22.5 months	21.5 months	21.0 months	21.0 – 22.5	21.75 months
Round 3	20.0 months	20.0 months	20.0 months	20.0 months	20.0 – 20.0	20.0 months